Sukhrob Ilyosbekov

Computer Vision · Deep Learning · Explainable AI

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Research Interests

Computer Vision, Deep Learning, Medical Image Analysis, Explainable AI, Object Detection and Segmentation

Education

Northeastern University, Boston, MA *January 2024 – May 2026*
- Master of Science in Computer Science; GPA: 4.0/4.0
- Relevant Coursework: CS 7180 Advanced Perception, Machine Learning, Computer Vision

Suffolk University, Boston, MA *September 2021 – May 2023*
- Bachelor of Science in Computer Science; Cum Laude; GPA: 3.5/4.0

INTI International College Subang, Malaysia *September 2019 – July 2021*
- Bachelor of Science in Computer Science (Transfer Program)

Tashkent University of Information Technologies, Uzbekistan *September 2017 – June 2019*
- Bachelor of Science in Software Engineering

Publications

[1] S. Ilyosbekov, S. Gajjar, and R. Jin, "MorphoCLIP: Text-Supervised Contrastive Learning for Perturbation Matching in Cell Painting Images," *arXiv preprint*, arXiv:2608.22690, 2026. [Paper]

[2] S. Ilyosbekov, "Localize, Don't Beautify: Client-Side Control of Image-Editing APIs for Cosmetic Surgery Previews," *arXiv preprint*, arXiv:2608.02841, 2026. [Paper]

[3] S. Ilyosbekov, "MelanomaNet: Explainable Deep Learning for Multi-Class Skin Lesion Classification," *arXiv preprint*, arXiv:2512.09289, 2025. [Paper]

Awards & Honors

- **National Physics Olympiad, 1st Place,** Uzbekistan; first student from my high school to win nationally
- **University Coding Competition Winner,** Tashkent University of Information Technologies
- **Outstanding Project Award,** Bookshelf Scanner, Computer Vision Course, Northeastern University

Technical Skills

Deep Learning & CV: PyTorch, TensorFlow/Keras, OpenCV, YOLO, EfficientNet, ResNet, U-Net, GradCAM++, CNNs, Vision Transformers, Image Classification, Object Detection, Depth Estimation, Super-Resolution, Explainable AI

Machine Learning: Supervised/Unsupervised Learning, Transfer Learning, XGBoost, K-means Clustering, Model Evaluation, Hyperparameter Tuning, NLP

Languages: Python, C#, C++, TypeScript, JavaScript, Kotlin

Infrastructure: Docker, Kubernetes, AWS, Azure, CUDA

Frameworks & Tools: .NET, Blazor, SignalR, Prisma, NestJS, ElysiaJS, Next.js, Angular, FastAPI, RabbitMQ, PostgreSQL

Research Projects

MorphoCLIP: Text-Supervised Cell Painting Retrieval GitHub | Paper
- Built a text-supervised contrastive model that matches Cell Painting microscopy images of drug- and gene-perturbed cells to natural-language treatment descriptions, pairing frozen vision/language backbones (DINOv3, BioClinical ModernBERT) with trainable projection heads so it trains on a single consumer GPU.
- Evaluated bidirectional image-to-text and text-to-image retrieval on the CPJUMP1 benchmark (51 plates, 3M+ cell images, 303 drugs, 160 genes) with integrated batch correction to remove plate-to-plate variation.
- **Stack:** Python, PyTorch, DINOv3, BioClinical ModernBERT, Contrastive Learning, CPJUMP1 Dataset

Localize, Don't Beautify: Client-Side Control of Image-Editing APIs for Cosmetic Surgery Previews Paper
- Studied why commercial image-editing APIs over-edit the face during cosmetic-surgery preview generation and how much of that can be fixed client-side without accessing the model.

- Compared prompt-only, masked-compositing, and model-based inpainting across six commercial editors and one inpainting model on facelift and rhinoplasty edits, scoring identity preservation (ArcFace) and localization accuracy on 196 edits; masked compositing beat model-based inpainting for localization.
- **Stack:** Image-Editing APIs, Inpainting Models, ArcFace, Prompt Engineering

MelanomaNet: Explainable Deep Learning for Multi-Class Skin Lesion Classification GitHub | Paper

- Developed an explainable deep learning system for multi-class skin lesion classification across all 9 ISIC 2019 diagnostic categories using EfficientNet V2 with high-resolution (384×384) input; achieved 85.61% accuracy and 0.8564 weighted F1 on 25,331 dermoscopic images.
- Implemented GradCAM++ attention visualization and novel ABCDE criterion analysis with automated feature extraction (asymmetry quantification, border irregularity, color variation via K-means, diameter calculation).
- Designed GradCAM-ABCDE alignment metrics to validate model attention correlation with clinical dermatological features, producing comprehensive interpretability reports.
- **Stack:** Python, PyTorch, EfficientNet V2, GradCAM++, OpenCV, Focal Loss, ISIC 2019 Dataset

LightDepth: Lightweight Monocular Depth Estimation GitHub

- Designed a lightweight depth estimation model using ResNet18 encoder with U-Net decoder and skip connections, achieving competitive accuracy with 42% fewer parameters (14.3M vs 24.8M) than Depth Anything V2.
- Achieved 72% faster inference while outperforming on relative error metrics (5.3% better AbsRel, 31.1% better SqRel) on NYU Depth V2.
- **Stack:** Python, PyTorch, ResNet18, U-Net Architecture, L1 Loss, NYU Depth V2 Dataset

FSRCNN: Accelerating Super-Resolution Convolutional Neural Network GitHub

- Implemented FSRCNN for single-image super-resolution (Dong et al., ECCV 2016), achieving 40× speedup over SRCNN with end-to-end learned upsampling at 2×, 3×, and 4× scales.
- Evaluated on Set5 (+1.78 dB PSNR, +0.0507 SSIM) and Set14 (+1.26 dB PSNR, +0.0477 SSIM) with ablation studies.
- **Stack:** Python, PyTorch, Mixed Precision Training (AMP), T91/Set5/Set14/DIV2K Datasets

Bookshelf Scanner: Multi-Modal Book Detection and Recognition GitHub

- Built an end-to-end pipeline combining YOLO segmentation for book instance detection with Moondream2 vision-language model for title/author extraction from bookshelf images.
- Integrated object segmentation with LLMs for multi-modal understanding and text recognition.
- **Stack:** Python, YOLO, Moondream2 VLM, llama.cpp, PyTorch, FastAPI, Angular

AoE4 Matchmaking: ML-Assisted Player Matching System GitHub

- Designed ML-assisted matchmaking combining unsupervised player clustering, XGBoost outcome prediction, and ELO rating for balanced competitive matches.
- Applied feature engineering on player statistics and ensemble methods to optimize match pairing.
- **Stack:** Python, XGBoost, K-means Clustering, Pandas, NumPy, FastAPI, Angular

Work Experience

Senior AI/ML Engineer | EmTech Care Labs | Portland, ME *Jan 2025 – Present*

- Built the machine-learning features for a HIPAA-compliant Alzheimer's care platform, including a risk model that flags patients showing early signs of decline from daily activity and care-log data so caregivers can step in sooner.
- Added an NLP pipeline that reads free-text clinical notes and extracts structured details (symptoms, medications, care events), and stood up the model-serving and monitoring infrastructure under HIPAA constraints.

Senior Machine Learning Engineer | AllFactors | Santa Clara, CA *Oct 2021 – Dec 2024*

- Built property-valuation and market-forecasting models on real-estate data that powered the product's analytics, turning raw pricing and inventory feeds into predicted values and trend signals.
- Designed the real-time feature pipeline that fed live market data to the models and pushed updated predictions to users, then productionized the models into the customer-facing dashboards.

Software Engineer | Smart Meal Service | Russia *Sep 2020 – Jul 2021*

- Built the software for a self-service food kiosk and robotic cashier in C# / .NET, plus the customer ordering app and admin dashboard, with automated order routing that sent each order straight to the right kitchen station.

Game Developer | Pentalight Technology | Malaysia *Mar 2020 – Feb 2021*

- Built the multiplayer networking for a VR smart-city simulation in Unity, syncing every participant's view in real time, plus spatial audio, hand-tracking, and a desktop spectator client.

Earlier: Freelance Developer at Freelancer.com (2019 – 2020), building Python NLP and .NET applications. Game Developer at EC Dev Team, Uzbekistan (2016 – 2019), led development of an RTS mod for Paradox's Clausewitz Engine.